

Selection of Feature-Sparse Ensembles: Co-Error, Certification, and Generalization

Research draft

Living development manuscript; not a submission-ready evaluation

Abstract

We investigate ensembles selected from large populations of feature-sparse weak predictors. The study asks whether empirical skill screening and explicit error complementarity improve generalization beyond random subspaces and ordinary validation-based ensemble selection. Squared-error and binary Brier losses admit an exact residual second-moment objective, whereas correlation-based diversity discards scale and can penalize useful negative dependence. The protocol reserves outer test folds, compares methods on shared candidate populations, and records selection and computational costs. An initial synthetic study finds modest gains over quality selection, severe screening infeasibility, and false survivors under the null, while strong forest and boosting baselines generally perform better. An oracle additive counterexample identifies attenuation from averaging subspaces. A follow-up affine diagnostic helps fixed subsets, but profiled selection improves OOF loss while usually worsening test loss. Candidate-pool scaling reveals conditional gains and increased selection optimism. Three real development tasks show gains over quality selection but a gap to simple linear baselines, conditional width effects, and duplicate-prediction confounds. An OOF-uniqueness control removes some selection gaps while worsening regression, implicating effective weighting rather than distinct independent specialists. A six-task public-data sweep trains 6,000 candidates per outer fold. Co-error gains over quality selection on five tasks, but loses molecule AUROC on Musk while improving row Brier. Its credit gain over the default forest is matched by a regularized forest, and boosting is stronger on every task. Pool and retained size have conditional, non-monotone effects. Results remain development evidence; no benchmark superiority or novelty claim is established.

1 Introduction

Useful tabular information may be distributed across weakly informative subspaces. Selecting complementary errors could improve an average of weak predictors, but this hypothesis competes with simpler explanations: regularization, more search compute, and ordinary ensemble selection. We test these alternatives using paired candidate pools and known synthetic mechanisms. Sparse signal and interactions provide explicit potential counterexamples.

2 Related Work and Unresolved Novelty

Bagging [Breiman, 1996a] and random forests [Breiman, 2001] reduce instability through aggregation. Random subspaces [Ho, 1998] assign fixed feature subsets to trees; random patches [Louppe and Geurts, 2012] also sample rows. RaSE [Tian and Feng, 2021] already generates and selects random subspaces, making a claim to novelty based on subspace search alone untenable. Ensemble selection [Caruana et al., 2004] optimizes validation ensemble performance from a model library. The ambiguity decomposition [Krogh and Vedelsby, 1995] motivates squared-loss aggregation; diversity can help or hurt majority votes [Brown and Kuncheva, 2010]. These precedents require differentiating any eventual contribution through mechanism, statistical procedure, scaling, or a substantiated algorithmic improvement. Quadratic pruning with cardinality constraints and SDP relaxations predates this study [Zhang et al., 2006]; the optimization formulation alone is not a contribution. Stacked regressions [Breiman, 1996b] fit nonnegative coefficients on

cross-validation predictions and explicitly formulate simplex weighting using residual-product quadratic loss. Our co-error matrix and scalar affine diagnostic therefore have close stacking precedents; restricting weights to a selected, equal-weight subset changes the constraint, not the loss identity. Portfolio weighting explicitly uses prediction-error covariances [Pérez-Rodríguez et al., 2023]. Stacking shrinkage theory for honest, nested least-squares subspaces [Chen et al., 2023] provides a useful regularization connection, but its assumptions do not cover our non-nested fitted sparse trees. Negative correlation learning jointly trains interacting networks [Liu and Yao, 1999], whereas our current models are generated independently and selected afterward. Random-projection ensembles also combine randomized search, validation selection, and aggregation [Cannings and Samworth, 2017]. Diversity penalties can change effective capacity [Reeve and Brown, 2018], motivating a regularization explanation. Cross-validation uncertainty is subtle under overlapping fits [Bengio and Grandvalet, 2004, Bates et al., 2024]. Portfolio-based weighting, conditional dependence, and selective inference require further full-text review.

Musk introduces a multiple-instance task: a molecule has many conformations and its label need not describe every conformation. This distinction is established in Dietterich et al. [1997]. Our max-pooled molecule score is a fixed diagnostic of inherited-label tree training, not a proposed multiple-instance algorithm. Entire-molecule holdouts are necessary to interpret generalization to new molecules.

3 Method

Candidate b has a fixed raw-feature mask F_b , tree depth, leaf-size constraint, row-sampling fraction, and seed, specified before fitting. Fold-local preprocessing and the tree see only F_b . Inner cross-fitting yields predictions P_{nb} . Binary screening uses an empirical bootstrap lower AUROC quantile, stratified by class for IID rows or resampling complete groups when configured; regression uses cross-fitted null-relative squared-error skill. We retain this as a heuristic screen, not a familywise certificate.

3.1 Theoretical Motivation

Let $E_{nb} = y_n - P_{nb}$ and $G = E^\top E/N$. For an equal-weight subset S of cardinality K ,

$$\hat{L}_{\text{sq}}(S) = K^{-2} \sum_{i,j \in S} G_{ij}. \quad (1)$$

This is an algebraic identity for regression MSE and binary Brier loss, not a generalization bound. Writing $\mu = N^{-1}E^\top \mathbf{1}$ gives $G = C + \mu\mu^\top$. Correlation removes marginal scale and mean; covariance alone removes bias. Neither equals squared ensemble loss. Statistical independence does not follow from zero correlation. Prediction correlation, error correlation, and useful ensemble complementarity are distinct.

For binary labels and valid probabilities, all residuals on a given example have the same sign: $y = 1$ gives $e_i \geq 0$ and $y = 0$ gives $e_i \leq 0$. Consequently $G_{ij} \geq 0$; negative uncentered co-error is impossible in this setting. Centered covariance/correlation does not have the same sign constraint. Even perfect residual correlation need not preclude a loss improvement through bias cancellation. On two equally weighted examples with $y = (0, 1)$, let $p_1 = (.7, .9)$ and $p_2 = (.1, .3)$. Both have Brier loss .25 and residual correlation 1, but their equal average has Brier loss .16: $G_{11} = G_{22} = .25$, $G_{12} = .07$, and residual means are $-.3$ and $.3$. This is an algebraic counterexample about squared loss, not empirical evidence of independence, new algorithmic novelty, or an AUROC gain.

For fixed fitted functions evaluated on a fresh observation, write $\eta(X) = \mathbb{E}[Y | X]$ and $\tau^2 = \mathbb{E}[\text{Var}(Y | X)]$. Conditional-mean orthogonality gives the population identity $\mathbb{E}[e_i e_j] = \tau^2 + \mathbb{E}[(\eta - p_i)(\eta - p_j)]$. The common noise term is constant under simplex aggregation. Bayes-optimal predictors can have identical, highly correlated residuals; forcing small raw correlation is therefore not a population optimality principle. Empirical OOF cross-terms and adaptive selection need not retain this orthogonality, motivating future synthetic controls using the known conditional mean to separate approximation complementarity from selection on realized noise.

For multiclass probabilities, define vector residuals $r_{ni} = u_{y_n} - p_{ni}$, where u_y is one-hot. We use $G_{ij} = N^{-1} \sum_n r_{ni}^\top r_{nj}$, giving sum-over-classes Brier loss. Classwise means μ_{ic} yield $G_{ij} = \sum_c C_{ij,c} + \sum_c \mu_{ic} \mu_{jc}$; covariance-only and shrinkage preserve this distinction. The initial multiclass correlation selectors apply

Pearson correlation to flattened sample/class residual coordinates. Because each probability vector sums to one, the flattened residual means are zero and this correlation is $G_{ij}/\sqrt{G_{ii}G_{jj}}$. It retains classwise bias and should not be interpreted as a correlation of class-centered residuals. This convention is recorded explicitly before interpreting the wine results.

We compare random selection, empirical quality, absolute and signed error correlation, minimax dependence, quality–dependence tradeoffs, and greedy co-error followed by improving one-swaps. Direct squared-loss selection without replacement is exactly the same greedy objective, so it supplies a verification baseline rather than a separate contribution. A with-replacement Caruana-style baseline yields nonuniform weights and is labeled separately. The loss-aligned quality-only control selects the K lowest diagonal elements G_{ii} , with stable candidate-ID ties and equal weights. It uses no cross-error terms. For regression it is equivalent to ranking individual cross-fitted skill on a common null denominator; for classification it tests whether comparing AUROC quality with Brier-based co-error confounds quality-criterion alignment.

3.2 Bias-Preserving Shrinkage Variant

We additionally test $G_\alpha = (1 - \alpha)C + \alpha \text{diag}(C) + \mu\mu^\top$. Shrinkage could reduce noisy cross-moment exploitation as the pool grows. It can also suppress genuine complementarity. The falsifying experiment holds K fixed, varies B/N , and measures outer loss and selection optimism. This regularizer has no established novelty or superiority claim.

3.3 An Oracle Attenuation Counterexample

Suppose $Y = \sum_{j=1}^d s_j(X_j) + \epsilon$, with independent centered signal components of equal variance v , independent centered noise of variance σ^2 , and population-optimal subspace learners $f_b = \mathbb{E}[Y \mid X_{F_b}] = \sum_{j \in F_b} s_j(X_j)$. Define $w_j = K^{-1} \sum_{b \in S} \mathbb{I}(j \in F_b)$. Then

$$L(S) = \sigma^2 + v \sum_{j=1}^d (1 - w_j)^2. \quad (2)$$

If each mask contains at most $q < d$ informative features, then $\sum_j w_j \leq q$ and Jensen’s inequality yields $L(S) \geq \sigma^2 + dv(1 - q/d)^2$. Thus averaging individually optimal subspace predictors attenuates distributed additive signal, even when their union covers all informative coordinates and B, K are arbitrarily large. For the additive regression development design, $d = 16$, $v = 1/16$, $q \leq 4$, and $\sigma^2 = 1$, so this idealized conditional-mean ensemble has risk at least 1.5625. This is a population calculation for these oracle learners, not a lower bound for arbitrary subspace functions, finite-sample fitted trees, or classification AUROC. Coordinated fitting or non-simplex rescaling can escape its assumptions. It motivates testing aggregate scale calibration separately from diversity.

3.4 Affine-Profiled Diagnostic Variant (APCE)

After Experiment 001, we test an affine correction $a\bar{P}_S + b$, with $a \geq 0$ and intercept b fitted exclusively on training OOF predictions. For a fixed subset, the profiled squared loss is

$$\widehat{L}_{\text{aff}}(S) = \widehat{\text{Var}}(Y) - \frac{\max(\widehat{\text{Cov}}(Y, \bar{P}_S), 0)^2}{\widehat{\text{Var}}(\bar{P}_S)}. \quad (3)$$

Zero aggregate variance gives the constant predictor. Centered prediction moments support incremental greedy and one-swap updates. The comparison includes affine correction of unchanged random, top-quality, and co-error subsets, isolating selection from the two new aggregation parameters. This is a stacking-like diagnostic and is not claimed as a novel optimizer or equal-output-weight primary method.

3.5 Observed OOF-Equivalence Control

Experiment 005 groups eligible candidates by exact equality of their training OOF vectors, retaining the first eligible ID per group before applying the unchanged random, quality, or co-error selector. It keeps K fixed

and reports infeasibility if too few groups remain. This is a mechanism control, not an assumed improvement or guarantee of functional/error independence. For an observed equivalence class c with common prediction vector P_c , an equal-specification subset is exactly $\bar{P}_S = \sum_c w_c P_c$ on the OOF matrix, where $w_c = |S \cap c|/K$. These are simplex weights in increments of $1/K$, bounded by the class multiplicity divided by K . Unique-subset selection instead constrains each selected class to weight $1/K$. This elementary weighting equivalence holds on observed OOF vectors; subsequent refits need not preserve it. No novelty claim is made for deduplication or this identity.

4 Experimental Protocol

All initial settings are fixed before each run. Outer folds remain unused during candidate screening and selection; selected specifications are refitted on the full outer training fold. Any future tuning regenerates the complete pipeline inside an additional tuning split, rather than slicing a globally fitted OOF matrix. All baselines share outer splits and fold-local preprocessing. Development synthetic regimes are separate from any locked real-world confirmation suite. Binary AUROC and regression RMSE are primary; multiclass log loss is primary. Brier loss is additionally central to the mechanism analysis. Bootstrap intervals over fixed OOF rows omit training instability, and unadjusted candidate screens omit multiplicity. These limitations prohibit interpreting them as simultaneous statistical certification. Shared candidate-search cost is charged to every standalone selector. Capacity and inference measurements accompany accuracy; matched-budget claims require dedicated experiments. Paired task-level analyses average folds and seeds before resampling tasks. No row-level significance claims across tasks are permitted.

4.1 Real Development Panel

Experiment 004 uses the complete sklearn-bundled Wisconsin diagnostic breast cancer and wine datasets [Wolberg et al., 1993, Aeberhard and Forina, 1992], and diabetes regression [Efron et al., 2004]. They contain respectively 569/30, 178/13, and 442/10 rows/features. All features are numerical and no entries are missing. These three tasks were named before scored results, to exercise binary, multiclass, and regression evaluation. They are development tasks, not a representative confirmation suite. IID row splitting is assumed; the bundled feature tables supply no grouping/time identifiers to evaluate that assumption.

The diabetes loader explicitly uses `scaled=False`; its default global centering/scaling would precede the evaluation splits. Any subsequent learned transform is fitted on training rows. The positive breast-cancer class is sklearn’s benign class, so its AUPRC describes that convention. Wine targets are encoded 0/1/2. Dataset versions, loader arguments, source attribution, raw data, content hashes, and target conventions accompany every run. The UCI sources state CC-BY-4.0; the diabetes source’s separate dataset license remains unspecified.

Two fixed configurations compare feature fractions .1 and .5 at $B = 100$, $K = 8$, depth 3 and minimum leaf size 5. With ceiling rounding, these produce 3/2/1 versus 15/7/5 features per learner across the three tasks. Seeds 11/29 and three outer and three OOF folds give six repeated evaluations per task/width; the same raw dataset is reused across seeds. Width changes the candidate population rather than generating nested feature masks. Unchanged baselines and outer/inner partitions must agree exactly across the two runs. Forest/boosting configurations are fixed, and a ridge ($\alpha = 1$) or logistic ($C = 1$) control uses fold-local standardization. No hyperparameter tuning is performed. Primary metrics remain task-specific; no pooled ranking or task-population interval is reported for this panel with one dataset per task type. Split effects are descriptive and their overlapping observations are not independent uncertainty replicates.

4.2 Larger Development Sweep (Registered Experiment 006)

A predeclared follow-up uses credit default [Yeh, 2009], the OpenML 23512 version-2 HIGGS subset [Whiteson, 2014, Baldi et al., 2014], MiniBooNE [Roe, 2005], superconductivity [Hamdicheh, 2018], and California housing [Pace and Barry, 1997]. Each is capped at 20,000 rows with fixed seed 7321, stratified for binary classification. Feature counts are 23, 28, 50, 81, and 8 respectively. The requested harder classification emphasis motivates the panel; observed baseline AUC is not an inclusion filter. California supplies a larger low-dimensional regression control. Exact repeated raw feature vectors are grouped in both outer and OOF

folds. This controls duplicate-row leakage, particularly in superconductivity, but supplies neither geographic nor unseen-composition extrapolation evidence. Public-source hashes, original row IDs, categorical conventions, source-count mismatches, licenses, and every scope restriction accompany the protocol.

Each outer fold generates 6,000 fixed-mask trees, balanced across fractions $\{.05, .20, .50\}$ and depths $\{3, 6\}$, minimum leaf size 10. Prefix pools $B = 1,000/3,000/6,000$ are crossed independently with $K = 16/64/128$ for random, quality, and co-error selection. Six homogeneous cells each have exactly $B = 1,000, K = 64$; additional controls include screening, correlation objectives, with-replacement selection, and one swap. The unscreened $B = 6,000, K = 64$ co-error setting is the predesignated anchor. One seed, three outer folds and two OOF folds are exploratory; no best outer-test setting is presented as a tuned model. Cluster-bootstrap multiplicities and matrix-free selector columns accelerate computations while matching literal-resampling and dense toy references. All settings are charged the complete shared-library search cost, a conservative accounting convention rather than a matched independent-training budget. The registered plan and smoke audit precede main scoring. All five original tasks and the molecule-level extension are complete.

A later extension, registered after credit default completed but before its own scores, adds Musk v2 [Chapman and Jain, 1994]: 6,598 conformations, 166 features, and 102 molecules. Both CV levels and bootstrap hold out/resample entire molecules; identifiers are excluded from every predictor. A fixed max over conformations after the row ensemble gives molecule AUROC, while row losses are also retained. Training and selection use inherited row labels and row Brier; they do not optimize the nonlinear molecule score, and the Gram identity is not claimed for that score. The extension is explicit development exploration, not independent confirmation.

An auxiliary control, registered before the main Musk run, fits a training-only scaled logistic model on $\log(1 + m)$, where m is the conformation count, with one observation per molecule on the unchanged outer partitions. It probes bag-size information without distance features. Under illustrative independent uniform row scores, the maximum of m scores has CDF t^m ; cardinality can therefore induce ranking without feature signal. These assumptions are not asserted for our fitted probabilities, and the count control cannot rule out all pooling effects.

Experiment 007 is registered after the first three classification results, before its own scores. It reuses all six immutable development datasets, specifications, OOF matrices and partitions at $B = 6,000, K = 64$, replacing individual classification AUROC quality by individual row squared loss. The selected specifications are refitted and evaluated once; regression supplies an expected unchanged-selection control. This separates criterion alignment from joint co-error selection, while remaining adaptive development reuse rather than fresh confirmation.

Reports preserve per-task outcomes, paired fold differences, resources and descriptive ranks among pre-designated references. Exploratory task-bootstrap intervals first average paired folds within tasks, then resample the four binary or two regression tasks; normalized squared loss avoids mixing regression units. These small chosen panels do not justify population coverage. Many comparisons are unadjusted exploratory estimates, without significance declarations.

The runner requests one BLAS thread, but NumPy uses Apple Accelerate on this host and threadpoolctl does not expose its thread counts. Actual numerical concurrency is therefore not asserted. Process CPU accounting includes all threads; matched wall-time budgets would require a separately verified runtime.

5 Synthetic Experiments

Experiment 001 evaluated nine known mechanisms for each of binary classification and regression: additive, redundant, correlated blocks, sparse, dominant, XOR, four-way parity, mixed, and null. Fixed settings were $N = 600, M = 40, B = 100, K = 8$, feature fraction 0.1, tree depth 3, minimum leaf size 5, two generation seeds, three outer folds, and three OOF folds. This produced 108 paired outer evaluations. Forest and boosting baselines use static configurations; search compute, final capacity, and inference budgets are measured but unmatched.

The initial hypothesis is not supported as a broad performance claim at these settings. Unfiltered co-error improves mean binary AUROC over unfiltered top quality by 0.0107, but generally trails the strong forests and boosting models. Co-error does not consistently improve over absolute residual correlation on outer

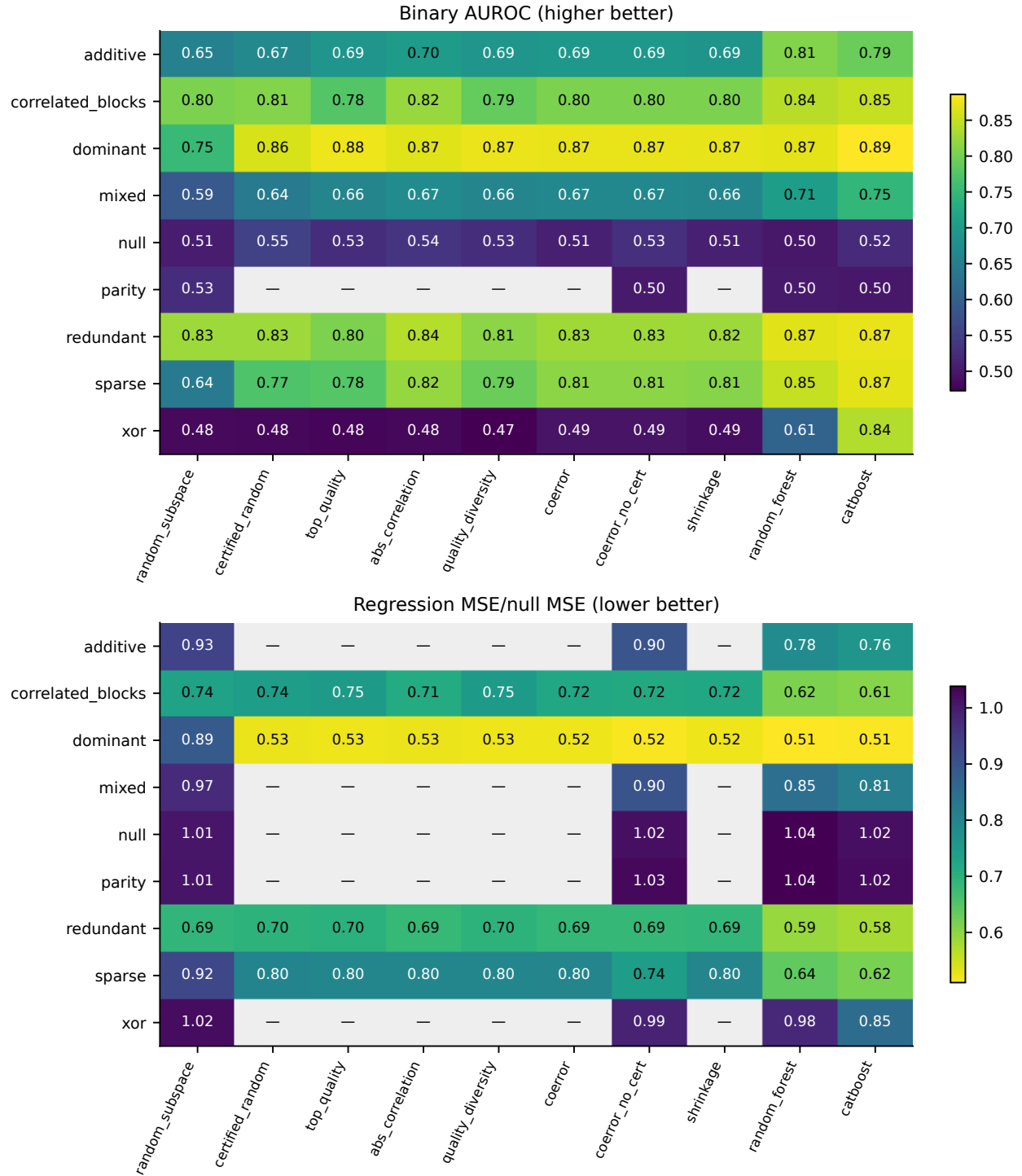


Figure 1: Development mechanism results. Gray cells indicate no feasible fixed- K screened ensemble. Numerical annotations are means of completed splits; the tables mark partial feasibility. Rankings are not confirmation evidence.

Regime	RS	Top	AbsCorr	CoErr	CoErr-all	RF	CatBoost
additive	0.655	0.689	0.701	0.693	0.693	0.809	0.794
correlated_blocks	0.805	0.775	0.822	0.804	0.804	0.842	0.849
dominant	0.752	0.875 [†]	0.867 [†]	0.872 [†]	0.872	0.869	0.886
mixed	0.588	0.662	0.672	0.666	0.666	0.705	0.746
null	0.506	0.527 [†]	0.535 [†]	0.509 [†]	0.529	0.500	0.522
parity	0.528	—	—	—	0.503	0.502	0.500
redundant	0.826	0.805	0.840	0.827	0.827	0.868	0.872
sparse	0.645	0.778	0.820	0.815	0.815	0.849	0.868
xor	0.483	0.478 [†]	0.481 [†]	0.494 [†]	0.495	0.605	0.838

Table 1: exp_001_mechanisms_v1: binary, auroc. Means over completed outer folds and seeds. —: fixed-K infeasible in every split; †: partial feasibility. Full coverage is stored separately. Static baselines, unmatched compute; development only.

Regime	RS	Top	AbsCorr	CoErr	CoErr-all	RF	CatBoost
additive	0.933	—	—	—	0.899	0.784	0.764
correlated_blocks	0.735	0.750 [†]	0.713 [†]	0.723 [†]	0.718	0.615	0.606
dominant	0.890	0.525 [†]	0.525 [†]	0.523 [†]	0.515	0.511	0.514
mixed	0.974	—	—	—	0.904	0.846	0.814
null	1.007	—	—	—	1.018	1.038	1.018
parity	1.009	—	—	—	1.027	1.039	1.023
redundant	0.691	0.701	0.686	0.687	0.687	0.591	0.580
sparse	0.925	0.797 [†]	0.795 [†]	0.797 [†]	0.744	0.639	0.617
xor	1.021	—	—	—	0.987	0.978	0.852

Table 2: exp_001_mechanisms_v1: regression, normalized_squared_loss. Means over completed outer folds and seeds. —: fixed-K infeasible in every split; †: partial feasibility. Full coverage is stored separately. Static baselines, unmatched compute; development only.

AUROC or squared loss; a positive aggregate Brier effect is concentrated in the dominant-feature regime. The $\alpha = 0.5$ shrinkage variant fails to improve on unregularized co-error in the compared task means. These are negative findings for the preregistered settings, not universal statements about the objectives.

5.1 Screening Failure and Feasibility

Screened co-error is feasible in only 40 of 54 binary splits and 16 of 54 regression splits. On every binary null split, 3–13 of 100 candidates pass the empirical screen, despite population AUROC 0.5 under this data-generating process. The bootstrap lower quantile must therefore remain labeled an empirical screen. Across shared feasible splits, top-quality screening has almost no incremental benefit over unscreened top quality; it frequently prevents a fixed-size ensemble.

5.2 Experiment 002: Attenuation Versus Selection

The follow-up fixes the candidate pools, masks, seeds, splits, and B/K settings from Experiment 001. It compares raw versus OOF-affine-calibrated random, top-quality, and co-error subsets, plus APCE, on six regression mechanisms. All subsets are unscreened to remove infeasibility as a confound. Each affine control selects exactly the same learners as its raw counterpart. No aggregation hyperparameter is tuned; these are two learned scalar parameters.

Affine calibration of co-error improves task-mean normalized loss in 5 of six regimes. Additive loss changes from 0.899 to 0.878, and mixed loss from 0.904 to 0.892. These improvements leave a large gap to the unrestricted forest on additive signal (0.784). The null control worsens from 1.018 to 1.031. Calibration therefore addresses part of the weakness without establishing a broadly competitive method or a unique attenuation explanation.

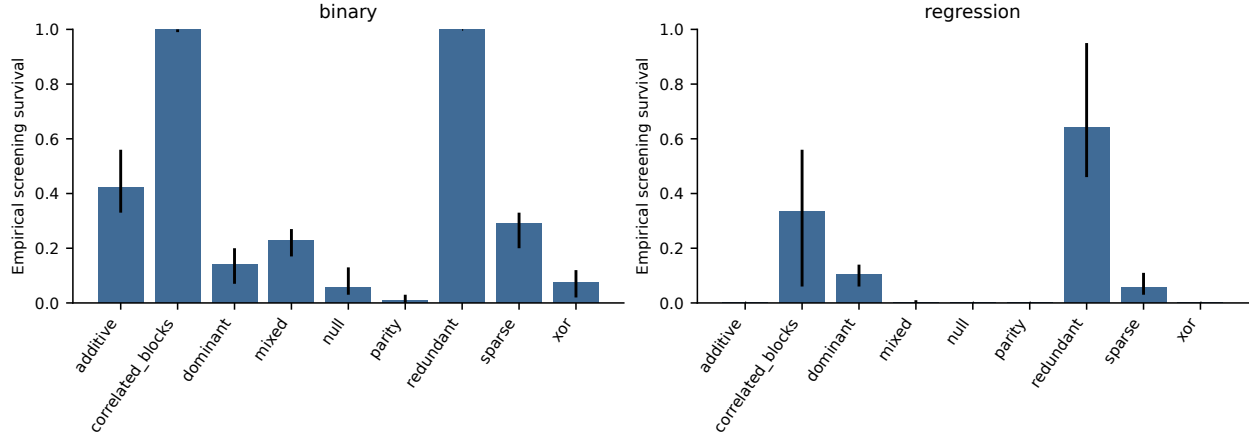


Figure 2: Empirical screening survival by regime. Error bars give the observed minimum and maximum over folds/seeds, not confidence intervals.

Regime	RS	RS+aff	Top	Top+aff	CoErr	CoErr+aff	APCE
additive	0.933	0.922	0.920	0.912	0.899	0.878	0.903
dominant	0.890	0.818	0.517	0.512	0.515	0.510	0.520
mixed	0.974	0.984	0.937	0.929	0.904	0.892	0.902
null	1.007	1.000	1.014	1.024	1.018	1.031	1.038
redundant	0.691	0.648	0.701	0.687	0.687	0.655	0.641
sparse	0.925	0.889	0.775	0.749	0.744	0.673	0.685

Table 3: Experiment 002: regression MSE divided by training-mean null MSE, averaged over six outer evaluations per regime. All subsets are unscreened. +aff fits two aggregate parameters on training OOF predictions; APCE also changes selection. B=100 and K=8 are fixed. Lower is better. Development evidence only.

APCE improves or ties calibrated OOF loss relative to affine co-error in all 36 splits, yet loses on outer normalized loss in 5 of six task means and wins only the redundant regime. On additive signal its outer loss is 0.903, versus 0.878 for calibration of the unchanged co-error subset. The additional optimization improves the reused training criterion while usually hurting generalization. This is consistent with selection optimism; smaller training fits for OOF predictions and the subsequent refit remain alternatives.

5.3 Experiment 003: Candidate Search at Fixed Ensemble Size

We vary $B \in \{30, 100, 300\}$ at fixed $K = 8$ on additive and null mechanisms for both tasks, holding all other settings fixed. Across the 48 adjacent-pool fold comparisons, masks, OOF predictions, and bootstrap columns agree exactly on their prefixes; the fixed forest also gives identical predictions. Larger pools improve binary additive co-error performance, while additive regression is non-monotone: normalized co-error loss is 0.908, 0.899, and 0.903, respectively.

The flexible profiled selector exposes increasing optimism. Its additive training OOF MSE decreases from 1.602 through 1.499 to 1.396, while outer normalized loss changes from 0.864 to 0.903 and 0.896. The endpoint deterioration occurs for both generation seeds in additive and null regression. Fixed $\alpha = 0.5$ shrinkage supplies no consistent outer remedy. Thus a larger library improves the reused criterion without a monotone generalization benefit for these selectors.

Binary null survivors rise from 0–5 at $B = 30$ to 7–39 at $B = 300$. Regression null screening retains no candidates at any pool size. This separates multiplicity-related false survival from the regression screen’s low feasibility/power. It does not estimate familywise coverage from six splits. All comparisons reuse development seeds and remain descriptive.

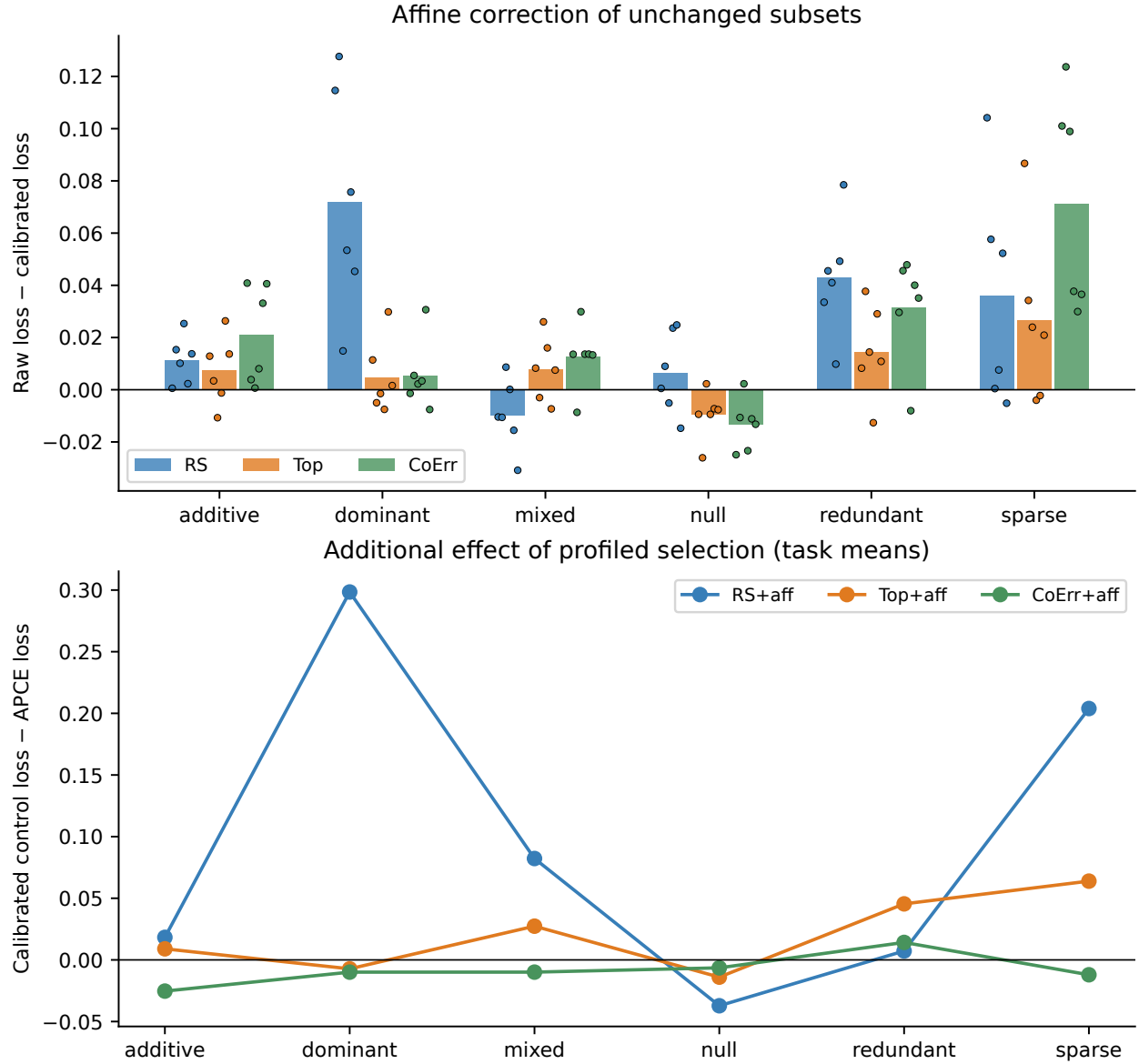


Figure 3: Affine diagnostic. Top: paired loss changes for unchanged subsets; dots show individual outer split effects and bars their means, not confidence intervals. Bottom: profiled selection relative to calibrated controls in task means. Positive values favor the added step. The null and most profiled co-error comparisons supply counterexamples.

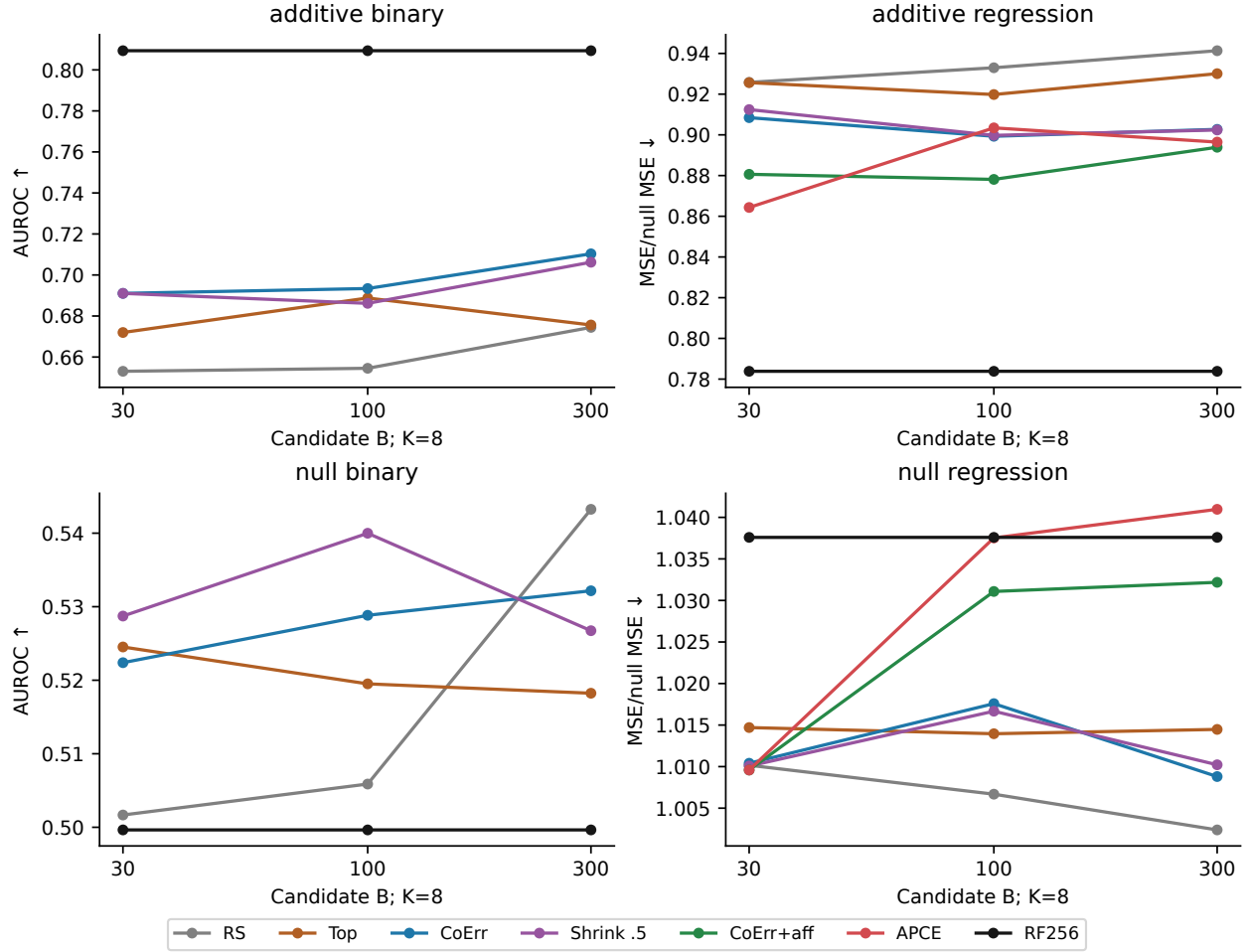


Figure 4: Candidate-pool scaling at fixed $K = 8$. Task means over six paired outer evaluations. Forest capacity is larger and budgets are unmatched; its unchanged predictions serve as a reference. Larger pools help binary additive co-error, but affine/profiled regression does not improve monotonically. Full split and generation-seed effects are stored with the artifacts.

6 Real-World Benchmarks

No confirmation benchmark has been evaluated. Current benchmark versions and baseline requirements will be recorded before locking the study.

6.1 Experiment 004: Three Real Development Tasks

The two preregistered feature-width runs complete all 36 outer evaluations; the paired-control audit verifies identical splits and unchanged baseline predictions. All task-applicable screened subsets are feasible. Empirical top-quality screening chooses the exact same subset and weights as unscreened top quality in all 36 paired cases. This panel supplies no evidence of incremental benefit from that screen for top-quality selection.

Unfiltered co-error improves primary task means relative to unfiltered top quality on all three datasets at both widths, but trails the fixed Random Forest reference in each case. The simple logistic/ridge control performs better still: breast-cancer AUROC is 0.995, wine log loss 0.079, and diabetes RMSE 54.974. The full table preserves stronger and weaker static baselines; this is not a tuned or resource-matched superiority comparison.

Increasing width changes wine co-error log loss from 0.231 to 0.161, approaching the forest (0.152) but remaining worse than logistic regression. Diabetes RMSE changes from 60.078 to 59.072, while the forest achieves 58.104. Breast-cancer AUROC worsens from 0.988 to 0.981. Wider masks are therefore not a universal remedy. Wine’s top-quality log loss worsens with width; stored probability-tail diagnostics reveal true-class zeros in three of six wide top-quality splits, making clipped log loss sensitive to those failures. Quality screening by macro AUROC does not control this behavior.

An exploratory duplicate-column audit exposes another explanation for selection gains. The diabetes narrow pool has exactly 10 distinct OOF prediction columns despite $B = 100$. Its eight top-quality specifications represent only 1–2 distinct observed prediction vectors; co-error represents 2–4. Equal weights over distinct specifications can thus produce unequal effective weights over repeated prediction vectors. OOF equality is not global functional identity and does not guarantee identical refits. This confound prevents attributing the quality-control gap solely to independent complementary learners. The follow-up ablation removes exact OOF duplicates before selection while retaining fixed K , comparing top quality, random, and co-error on the same eligible pool.

The selected tree ensembles are smaller in serialized bytes than the 256-tree forest, but require more charged search CPU time. Pipeline/preprocessing overhead also limits inference savings relative to tree-count differences. Stored resource measurements and the compute–performance figure preserve these tradeoffs; no isolated-process memory or matched-budget advantage is established.

6.2 Experiment 005: Duplicate Avoidance Versus Implicit Weighting

All 18 follow-up folds reuse the narrow Experiment 004 pools exactly; the audit verifies identical raw data, specifications, splits, OOF and bootstrap arrays, and 108 unchanged-control prediction pairs. All 54 OOF-unique subsets contain eight exact-distinct training prediction vectors with equal weights. Screening is removed as a confound and all applicable variants are feasible.

On diabetes, deduplication worsens co-error RMSE from 60.078 to 65.978; unique top quality achieves 65.945. The original co-error advantage over quality disappears while both unique methods remain weak. Removing duplicate concentration is therefore not a remedy: forcing eight observed patterns prevents concentrating weight on a few useful univariate predictors. The original co-error closely matches the frequency-weighted Caruana reference. This supports implicit weighting as an explanation for much of the diabetes quality-control gap, rather than eight distinct complementary specialists. It does not identify a unique causal mechanism beyond the intervention, because representative/refit changes and the hard uniqueness constraint alter the method.

Wine’s unique top-quality and co-error log losses are both approximately 0.237, versus 0.248 and 0.231 originally. The primary-metric gap nearly vanishes; a modest Brier advantage remains in the stored secondary metrics. Cancer top quality and co-error are unchanged in the reported primary and Brier split effects, consistent with its already mostly unique pool. Thus exact duplicate structure changes the interpretation

Method	Cancer AUROC $\uparrow$		Wine log loss $\downarrow$		Diabetes RMSE $\downarrow$	
	.1	.5	.1	.5	.1	.5
Random subspace	0.984	0.980	0.457	0.169	67.498	59.023
Screened random	0.984	0.980	0.457	0.169	63.594	59.506
Screened top quality	0.981	0.974	0.248	0.502	64.453	59.980
Top quality (unfiltered)	0.981	0.974	0.248	0.502	64.453	59.980
Absolute residual correlation	0.990	0.986	0.294	0.176	60.695	59.459
Signed residual correlation	0.990	0.986	0.294	0.176	60.695	59.459
Minimax residual correlation	0.989	0.984	0.320	0.176	61.502	59.156
Quality + diversity	0.988	0.985	0.231	0.165	61.071	59.642
Prediction correlation	0.957	0.985	0.612	0.233	61.719	60.628
Co-error greedy	0.988	0.982	0.228	0.162	59.826	58.825
Direct squared loss	0.988	0.982	0.228	0.162	59.826	58.825
Screened co-error + swaps	0.988	0.981	0.231	0.161	59.826	59.072
Co-error (unfiltered)	0.988	0.981	0.231	0.161	60.078	59.072
Covariance only	0.988	0.981	0.234	0.162	59.826	59.072
Shrinkage .5	0.987	0.981	0.227	0.167	59.791	58.783
Caruana (frequency weights)	0.986	0.981	0.225	0.163	60.077	59.042
Direct AUROC	0.991	0.979	–	–	–	–
Co-error + affine	–	–	–	–	58.458	58.554
APCE	–	–	–	–	58.277	58.932
Shallow Random Forest	0.984	0.984	0.255	0.255	57.098	57.098
Shallow ExtraTrees	0.983	0.983	0.476	0.476	58.337	58.337
Random Patches	0.986	0.982	0.426	0.246	68.739	59.161
Random Forest (256)	0.989	0.989	0.152	0.152	58.104	58.104
ExtraTrees (256)	0.994	0.994	0.158	0.158	57.315	57.315
XGBoost	0.991	0.991	0.140	0.140	59.493	59.493
LightGBM	0.991	0.991	0.118	0.118	58.589	58.589
CatBoost	0.993	0.993	0.168	0.168	55.902	55.902
Logistic / ridge	0.995	0.995	0.079	0.079	54.974	54.974

Table 4: Experiment 004: three real development datasets, B=100 and K=8 for subset methods, depth 3, six outer evaluations per dataset/width. Caruana uses eight steps and can retain fewer unique learners. Each pair of columns changes only feature fraction (.1/.5) and its Random Patches control. Means use completed splits; $\uparrow$ marks partial feasibility, – indicates infeasible or task-restricted. Baselines use static settings and unmatched resources. Wine co-error optimizes Brier, not its primary log loss. Repeated CV folds are not independent datasets.

Method	Cancer AUROC $\uparrow$	Wine log loss $\downarrow$	Diabetes RMSE $\downarrow$
Random subspace	0.984	0.457	67.498
Top quality (unfiltered)	0.981	0.248	64.453
Co-error (unfiltered)	0.988	0.231	60.078
Random subspace (OOF-unique)	0.986	0.411	67.536
Top quality (unfiltered) (OOF-unique)	0.981	0.237	65.945
Co-error (unfiltered) (OOF-unique)	0.988	0.237	65.978
Caruana (frequency weights)	0.986	0.225	60.077
Random Forest (256)	0.989	0.152	58.104
Logistic / ridge	0.995	0.079	54.974

Table 5: Experiment 005: exact training-OOF duplicate removal at fixed $K=8$, $B=100$ and feature fraction .1. Original and OOF-unique selectors share all data, specifications, OOF predictions and splits; unique variants retain first-ID representatives. Six overlapping outer evaluations per task, unscreened. Caruana has eight steps and frequency weights. OOF equality does not imply global function equality. Static forest/linear references have unmatched budgets.

Task	Study rows	Features	Split groups	Missing cells
Credit default	20,000	23	19,972	0
HIGGS	20,000	28	19,999	0
MiniBooNE	20,000	50	19,917	0
Superconductivity	20,000	81	14,397	0
California housing	20,000	8	20,000	0
Musk v2 (molecules)	6,598	166	102	0

Table 6: Stored study samples, distinct from full-source catalog counts. Exact raw feature-vector groups for five tasks; molecule groups for Musk. Missing cells are imputed inside training folds. MiniBooNE’s retained –999 sentinels do not count as missing cells.

of selection gains in some regimes, but cannot explain every gain. No significance claim follows from these three reused development tasks.

6.3 Experiment 006: Larger Public-Data Sweep

All six registered tasks are complete: 1,098 test-prediction records, three outer folds per task, and 6,000 candidates per fold. The two-fold OOF libraries require 216,000 base fits, before selected refits and baseline fits. Artifact audits reconstruct metrics and row squared-loss identities, with maximum discrepancy below 2.9×10^{-13} . All 52 selection settings and nine static baselines per task are retained; the anchor is not a test-selected winner.

Credit default lies in the requested harder baseline region: the fixed forest achieves AUROC 0.767 and logistic regression 0.728. The predesignated co-error anchor reaches 0.779, exceeding top quality (0.776) and the default forest, but trailing CatBoost. The predeclared forest with minimum leaf size five reaches .779, nearly identical to co-error at displayed precision. The apparent default-forest gain is therefore not a gain over the forest family or evidence beyond ordinary regularization. On HIGGS, co-error improves over top quality (0.764 versus 0.746) yet trails the forest (0.785) and all three boosting references. MiniBooNE is easier for these baselines than the requested AUC region; it remains in the panel. Its anchor also trails the forest and boosting references despite improving over top quality. These conditional observations establish neither broad superiority nor an independent diversity mechanism.

Increasing the nested library from $B = 3,000$ to 6,000 at fixed $K = 64$ slightly worsens co-error task-mean AUROC on both credit default and HIGGS. Thus more candidate search does not monotonically improve generalization even on these larger tasks. MiniBooNE has a small endpoint improvement. The regression tasks show modest improvement across $B = 1,000/3,000/6,000$, while increasing K beyond 64 worsens both regression task means. These curves refute a universal monotone benefit from either search size or retained size.

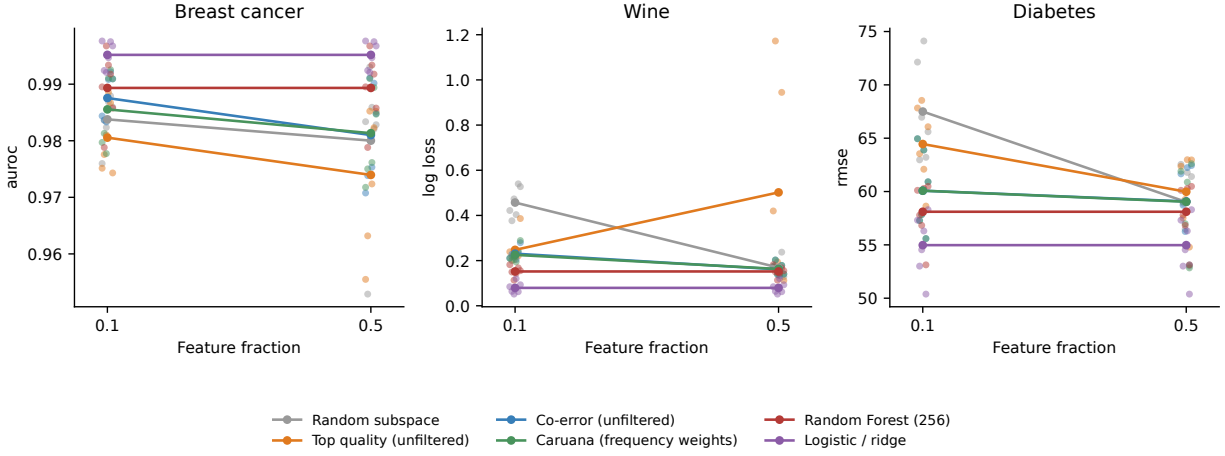


Figure 5: Real development feature-width comparison. Lines are task means and dots are overlapping outer split observations, not confidence intervals. The three primary metrics retain their own scales. Linear controls outperform these selected sparse-tree ensembles; larger masks help conditionally.

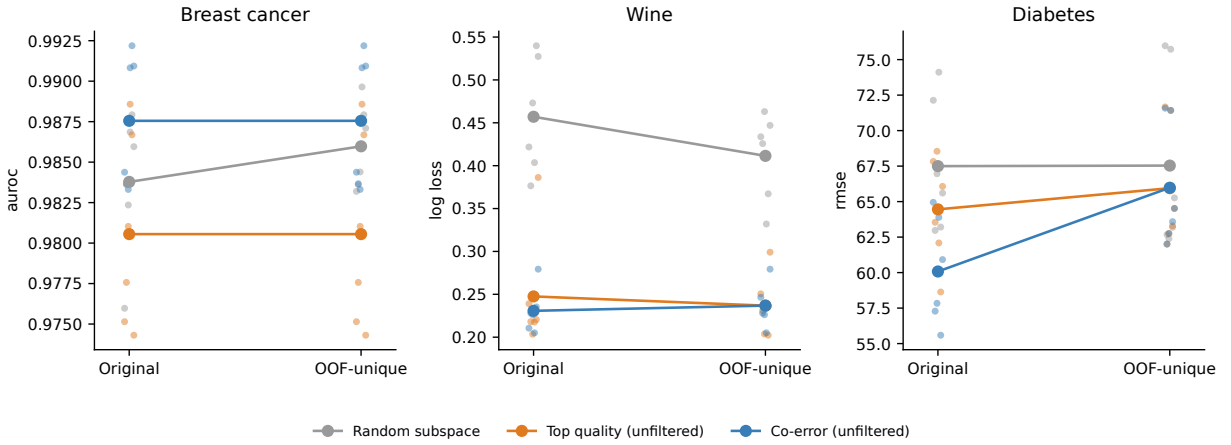


Figure 6: Exact OOF-equivalence intervention at fixed $K = 8$. Task means and overlapping split observations. Co-error's diabetes improvement depends on allowing repeated observed patterns; restricting to eight unique patterns hurts generalization. Uniqueness is neither statistical independence nor a universal algorithmic improvement.

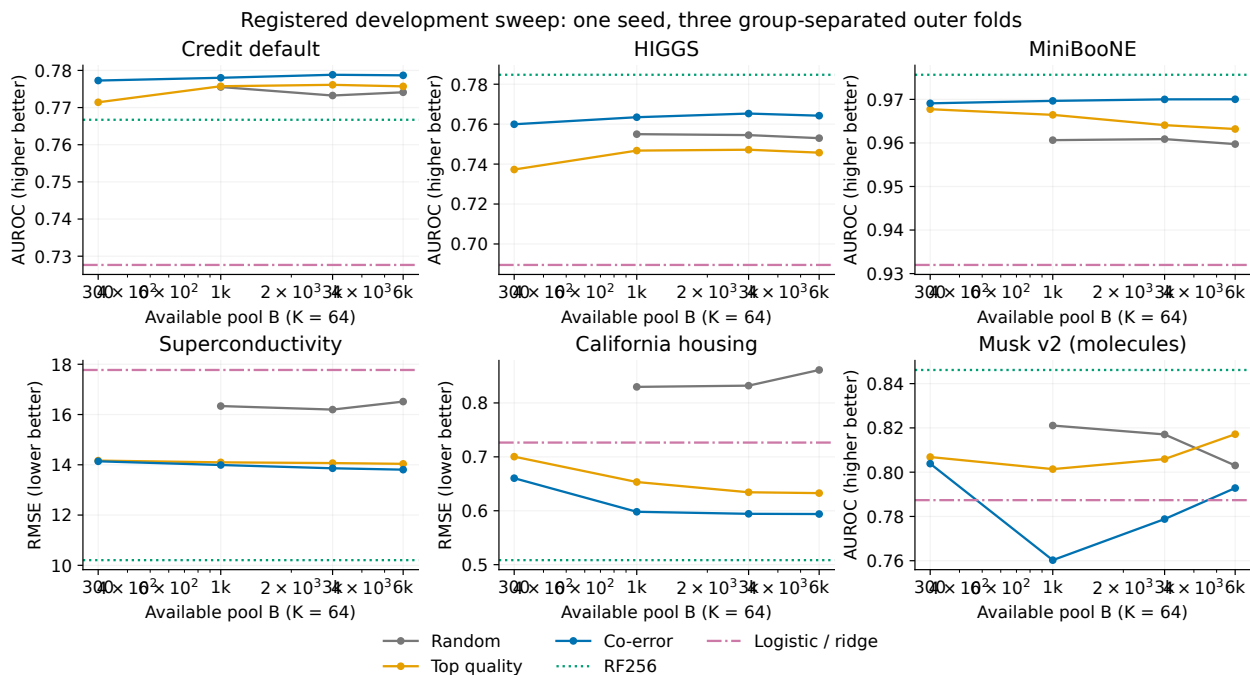


Figure 7: Nested available-pool size at fixed $K = 64$; task means over three paired development folds. All settings share the full generated library and are charged its cost, so the horizontal axis is not a matched independent training budget. Musk uses molecule-max AUROC; regression uses RMSE. No confidence bars are implied.

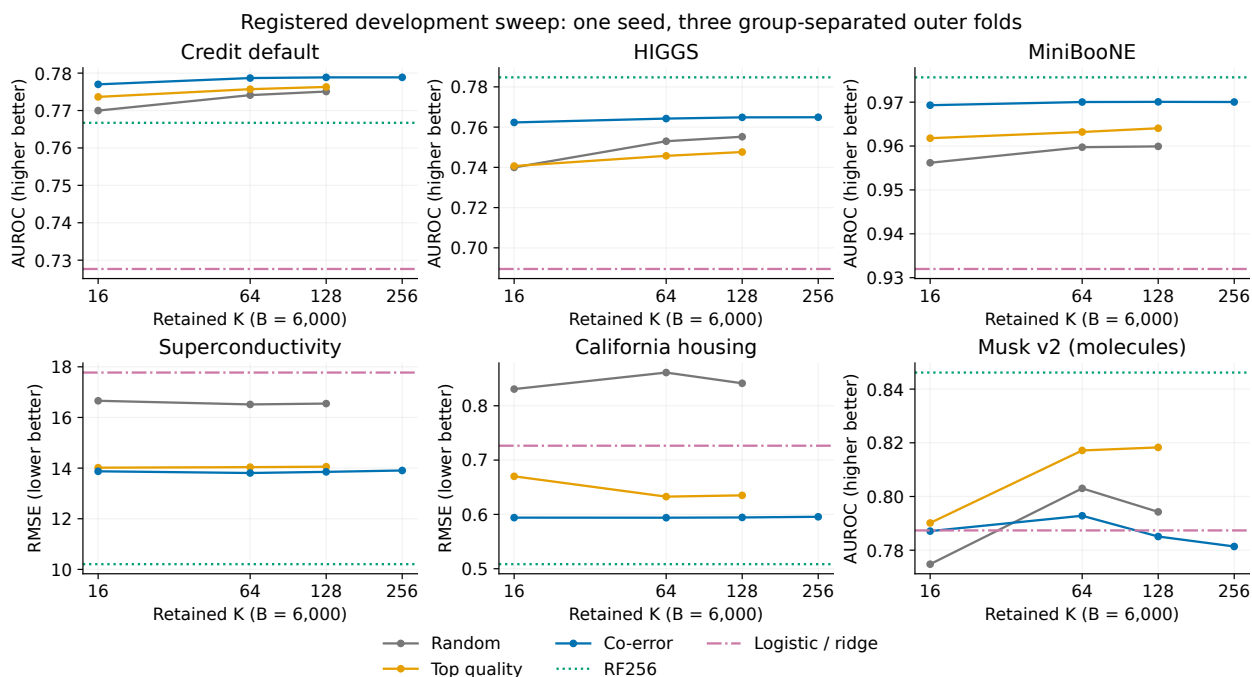


Figure 8: Retained size varies independently of generated $B = 6,000$. Larger ensembles are not uniformly better. Fixed forests and linear references have different capacities and unmatched costs.

Task / metric	Co-error	Top	Caruana	RF	RF leaf5	ET	XGB	LGB	Cat	Linear
Credit default / AUC	0.779	0.776	0.778	0.767	0.779	0.746	0.776	0.775	0.784	0.728
HIGGS / AUC	0.764	0.746	0.764	0.785	0.786	0.767	0.792	0.794	0.793	0.689
MiniBooNE / AUC	0.970	0.963	0.970	0.976	0.975	0.976	0.982	0.982	0.981	0.932
Superconductivity / RMSE	13.807	14.038	13.840	10.206	10.342	9.899	10.246	10.324	11.641	17.773
California housing / RMSE	0.594	0.633	0.594	0.508	0.516	0.505	0.466	0.460	0.489	0.726
Musk v2 (molecules) / AUC	0.793	0.817	0.807	0.846	0.842	0.872	0.868	0.885	0.864	0.787

Table 7: Predesignated $B = 6,000$, $K = 64$ co-error anchor and static references. Means of three outer folds and one seed. Exact feature-vector groups for five tasks; molecule groups and molecule-max AUC for Musk. Other sweep cells are reported separately; no test-based winner selection or resource matching.

Co-error improves over quality selection on both regressions, but remains weaker than the forest: superconductivity RMSE is 13.807 versus 14.038 and 10.206; California RMSE is 0.594 versus 0.633 and 0.508. With-replacement Caruana selection is nearly identical on California (0.594) and close on the original three classifiers. These results provide no substantial new advantage over ordinary ensemble selection.

6.4 Molecule-Level Counterexample and Pooling Control

Musk co-error molecule AUROC is 0.793, below top quality (0.817), the forest (0.846) and all boosting references. Row Brier nevertheless improves from .131 for quality to .123 for co-error; Caruana reaches .115 and molecule AUROC 0.807. Thus a row squared-loss improvement need not transfer to a nonlinear molecule ranking metric. The count-only control averages molecule AUROC .454, with fold values .498/.456/.408. It does not support a simple count-only explanation, but cannot exclude interactions between score distributions, cardinality and max pooling. There are only 102 independent-molecule units under the assumed sampling model, not 6,598 independent conformations. This is a negative primary result.

6.5 Resources and Objective Diagnostics

Compute, capacity, raw serialized size and row inference measurements accompany every method. Search costs include the entire shared library; raw model bytes include fit-row provenance, and phase RSS includes shared arrays and other models. The resource plots are measured points with static settings and unmatched budgets, not optimized frontiers or evidence of budget-matched superiority.

On all nine original classification folds, screening leaves both co-error and top-quality IDs, weights and predictions exactly unchanged. Absolute and signed residual-correlation selectors choose the same subsets on all 18 panel folds; this panel cannot discriminate those notions by comparing their identical outputs. On HIGGS and MiniBooNE correlation selection improves AUROC relative to co-error but worsens Brier, further separating ranking and squared-loss objectives. Unlike the narrow diabetes experiment, both quality and co-error select 64 exact distinct OOF columns on every larger-panel fold. Exact repeated selected columns therefore do not explain this panel’s gap, although correlated or nearly equal functions and effective concentration remain possible. The full pools have approximately 3,650–6,000 distinct OOF patterns; specification count is not a count of statistically independent learners. Musk screening survival also varies sharply across folds (1,372–5,235 of 6,000), reinforcing the sampling-unit caveat.

7 Ablations and Analysis

The stored ablations include all required v0 components, signed versus absolute residual correlation, prediction correlation, covariance without bias, direct squared-loss greedy selection, with-replacement selection, and shrinkage. An artifact audit reconstructed loss and test metrics for all selected methods; the maximum squared-loss/co-error identity discrepancy was below 10^{-10} , including the larger-scale raw diabetes targets. Direct squared-loss and co-error greedy validation objectives agree, confirming that their distinction supplies no additional mechanism.

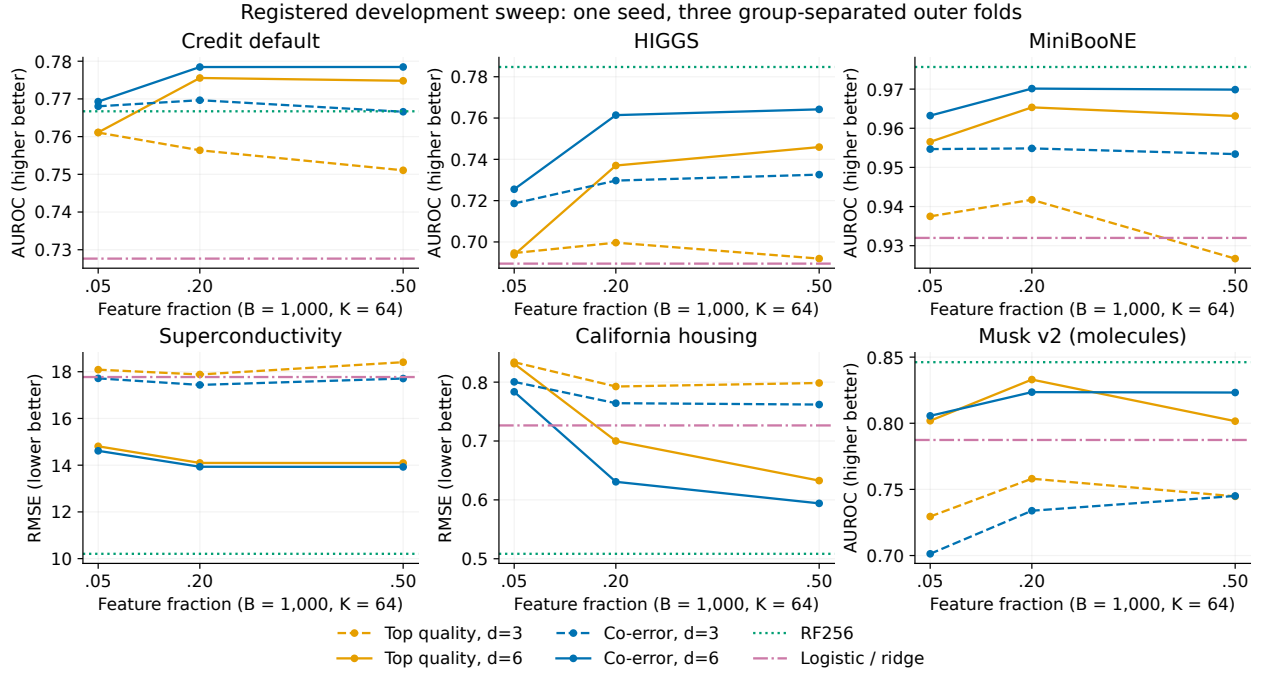


Figure 9: Homogeneous feature-fraction/depth cells at matched available $B = 1,000$ and $K = 64$. Width and depth alter representation and individual strength. The cells are descriptive ablations, not test-tuned hyperparameters.

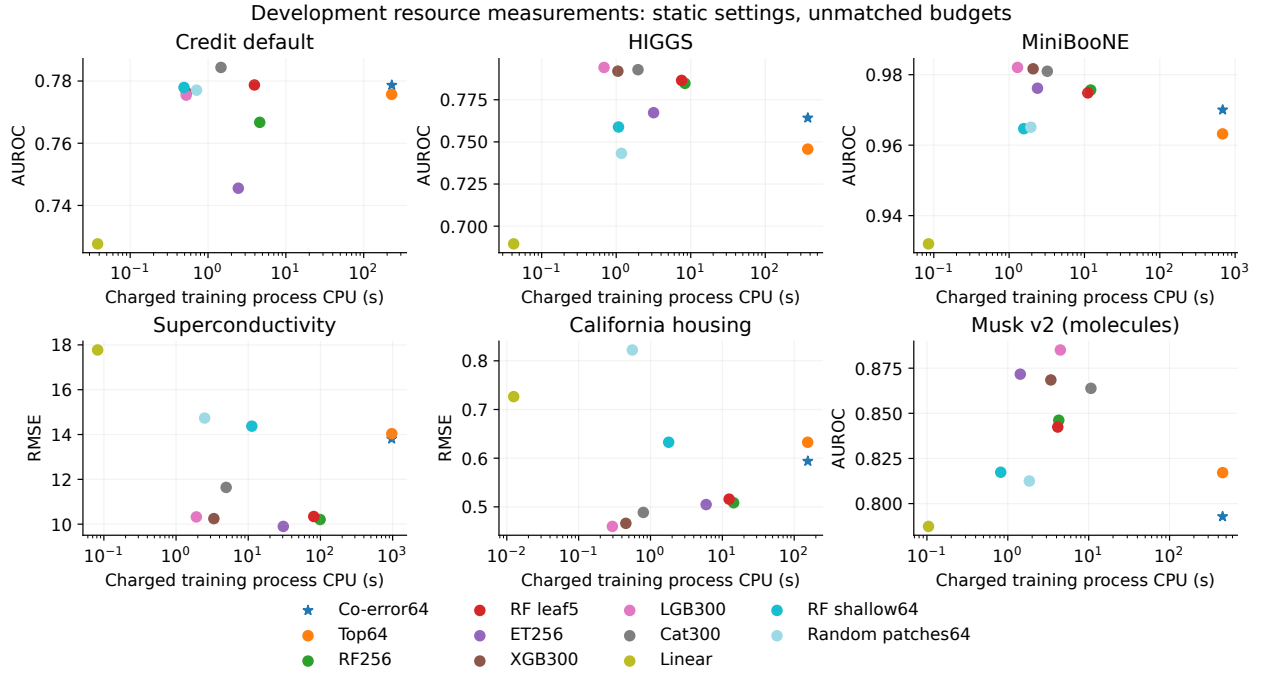


Figure 10: Performance versus charged process CPU. Selected methods pay for OOF search and screening plus selection/refit; static references have no comparable hyperparameter search. Process CPU includes numerical threads.

8 Analysis and Next Discriminating Test

OOF/test loss gaps combine selection, training-size, refit and sampling effects; they do not uniquely identify optimism. Independent selection holdouts must compare retained fitted functions with refitted specifications. A conservative bounded-differences bound can simultaneously certify binary AUROC for a fixed independent library [McDiarmid, 1989], with no automatic guarantee for OOF scores or refits. The duplicate-control result motivates comparing regularized simplex weights on canonical prediction patterns with multiplicity-constrained discrete weights. Additive synthetic linear/oracle controls remain necessary to separate representation mismatch from selection failure. These tests precede tuned, resource-matched confirmation.

9 Limitations

This draft does not establish a novel algorithm or strong benchmark performance. OOF selection risks overfitting to a reused training prediction matrix. Refitting changes the fitted functions and may change their error dependence. IID row bootstrap screens do not justify guarantees under overlapping training folds, large candidate libraries, grouped observations, or temporal shifts. Pooled OOF AUROC compares outputs from different fitted functions and can be distorted by calibration/prevalence shifts: in a constructed two-group example with fold positive rates $1/4$ and $3/4$, cross-fitted null predictions give pooled AUROC .25 while each held-out fold gives .5. Outer-test metrics use one fixed refit per fold; the current pooled OOF quality/screen remains an explicit estimation limitation. Independent fixed-function validation and same-fit ranking controls are future work. Sparse shallow trees can fail on interactions that need joint feature access. Only two generation seeds and one dimensional/sample-size setting have been tested in the completed synthetic studies. The initial three real tasks are small numerical development datasets; the six-task public-data sweep adds only one seed and chosen tasks. Musk contributes 102 molecules with a nonlinear pooling primary metric different from its row selection objective. Repeated folds of the same rows are not independent scientific replicates. Exact-vector groups control duplicate leakage but do not establish temporal, geographic, or chemical-scaffold generalization. Duplicate OOF columns can make equal specification weights into effective frequency weights, and flattened multiclass correlation includes classwise bias. Static baselines lack tuning and resources are unmatched. Phase RSS includes the runner’s retained models and shared arrays, so it does not estimate isolated model memory. Synthetic regimes can share generation randomness; descriptive task resampling does not supply calibrated interval coverage. Linear controls have been tested on the real panel; additive synthetic linear/oracle controls and competitive tuned modern methods remain untested. Partial screening feasibility changes the support of comparisons; aggregate effects can be driven by one mechanism. These restrictions preclude real-world or broad algorithm-superiority claims.

10 Conclusion

The development studies do not establish a broadly competitive or novel method. Unadjusted OOF screening can fail, equal averaging can attenuate additive signal, and co-error gains over quality are conditional. Affine calibration helps some fixed subsets, while flexible selection and larger pools can increase optimism. The initial small real panel favors linear controls; the larger classification panel favors boosting references; a regularized forest matches the apparent credit gain over the default forest. Musk supplies a primary-metric counterexample to gains over quality selection, despite improved row Brier. Exact OOF deduplication eliminates some quality-selection gaps while hurting regression, implicating useful implicit weighting rather than independent specialists. Independent selection/refit and representation controls, tuned strong baselines, and resource-matched confirmation remain necessary.

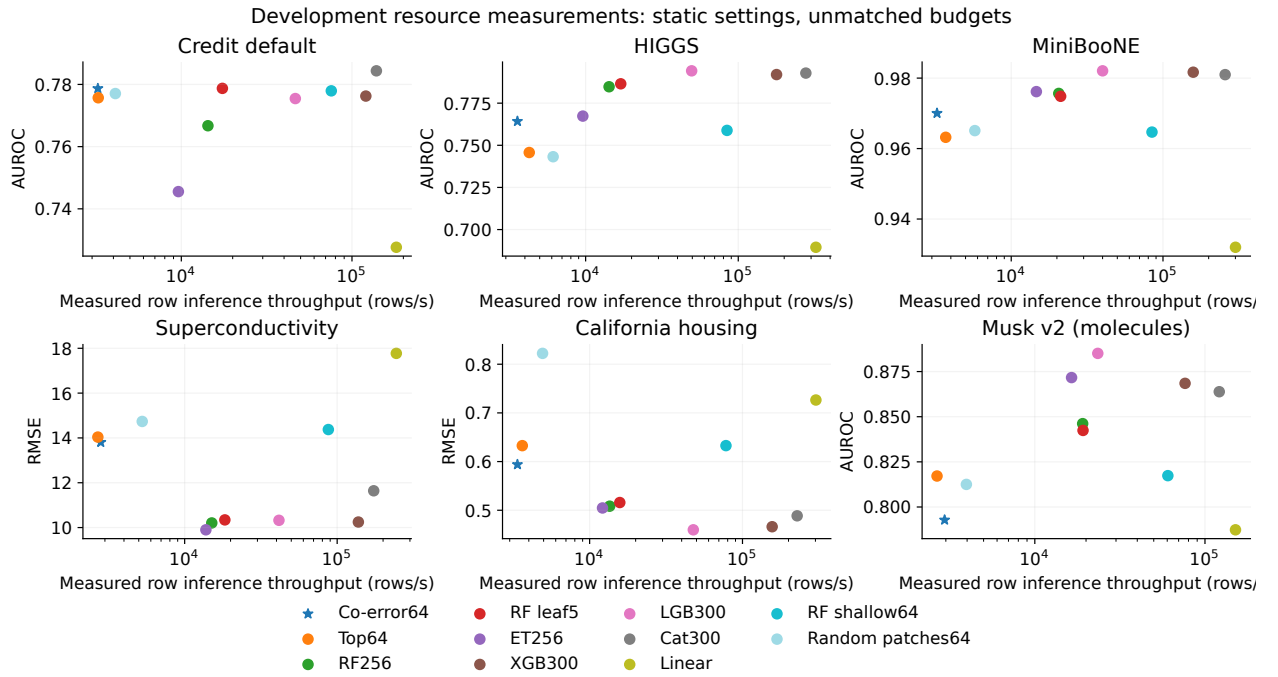


Figure 11: Measured row inference throughput on training inputs, including each learner's preprocessing. Musk molecule aggregation is not included in row throughput. Timings are development measurements on one host.

References

- Stefan Aeberhard and M. Forina. Wine, 1992. URL <https://archive.ics.uci.edu/dataset/109/wine>.
- Pierre Baldi, Peter Sadowski, and Daniel Whiteson. Searching for exotic particles in high-energy physics with deep learning. *Nature Communications*, 5:4308, 2014. doi: 10.1038/ncomms5308.
- Stephen Bates, Trevor Hastie, and Robert Tibshirani. Cross-validation: What does it estimate and how well does it do it? *Journal of the American Statistical Association*, 2024. doi: 10.1080/01621459.2023.2197686.
- Yoshua Bengio and Yves Grandvalet. No unbiased estimator of the variance of k-fold cross-validation. *Journal of Machine Learning Research*, 5:1089–1105, 2004. URL <https://jmlr.org/papers/v5/grandvalet04a.html>.
- Leo Breiman. Bagging predictors. *Machine Learning*, 24:123–140, 1996a. doi: 10.1007/BF00058655.
- Leo Breiman. Stacked regressions. *Machine Learning*, 24:49–64, 1996b. doi: 10.1007/BF00117832. URL <https://statistics.berkeley.edu/sites/default/files/tech-reports/367.pdf>.
- Leo Breiman. Random forests. *Machine Learning*, 45:5–32, 2001. doi: 10.1023/A:1010933404324.
- Gavin Brown and Ludmila I. Kuncheva. Good and bad diversity in majority vote ensembles. In *Multiple Classifier Systems*, pages 124–133, 2010. doi: 10.1007/978-3-642-12127-2_13.
- Timothy I. Cannings and Richard J. Samworth. Random-projection ensemble classification. *Journal of the Royal Statistical Society: Series B*, 79(4):959–1035, 2017. doi: 10.1111/rssb.12228.
- Rich Caruana, Alexandru Niculescu-Mizil, Geoff Crew, and Alex Ksikes. Ensemble selection from libraries of models. In *ICML*, 2004. doi: 10.1145/1015330.1015432.
- David Chapman and Ajay Jain. Musk (version 2). UCI Machine Learning Repository, 1994.
- Xin Chen, Jason M. Klusowski, and Yan Shuo Tan. Error reduction from stacked regressions. *arXiv preprint arXiv:2309.09880*, 2023. URL <https://arxiv.org/html/2309.09880v3>.
- Thomas G. Dietterich, Richard H. Lathrop, and Tomás Lozano-Pérez. Solving the multiple instance problem with axis-parallel rectangles. *Artificial Intelligence*, 89(1–2):31–71, 1997. doi: 10.1016/S0004-3702(96)00034-3.
- Bradley Efron, Trevor Hastie, Iain Johnstone, and Robert Tibshirani. Least angle regression. *The Annals of Statistics*, 32(2):407–499, 2004. doi: 10.1214/009053604000000067. URL https://web.stanford.edu/~hastie/Papers/LARS/LeastAngle_2002.pdf.
- Kam Hamidieh. Superconductivity data. UCI Machine Learning Repository, 2018.
- Tin Kam Ho. The random subspace method for constructing decision forests. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(8):832–844, 1998. doi: 10.1109/34.709601.
- Anders Krogh and Jesper Vedelsby. Neural network ensembles, cross validation, and active learning. In *Advances in Neural Information Processing Systems*, volume 7, 1995.
- Yong Liu and Xin Yao. Ensemble learning via negative correlation. *Neural Networks*, 12(10):1399–1404, 1999. doi: 10.1016/S0893-6080(99)00073-8.
- Gilles Louppe and Pierre Geurts. Ensembles on random patches. In *ECML PKDD*, pages 346–361, 2012. doi: 10.1007/978-3-642-33460-3_28.
- Colin McDiarmid. On the method of bounded differences. In J. Siemons, editor, *Surveys in Combinatorics, 1989*, pages 148–188. Cambridge University Press, 1989. doi: 10.1017/CBO9781107359949.008.
- R. Kelley Pace and Ronald Barry. Sparse spatial autoregressions. *Statistics and Probability Letters*, 33(3):291–297, 1997. doi: 10.1016/S0167-7152(96)00140-X.
- Javier Pérez-Rodríguez, Francisco Fernández-Navarro, and Thomas Ashley. Estimating ensemble weights for bagging regressors based on the mean–variance portfolio framework. *Expert Systems with Applications*, 229:120462, 2023. doi: 10.1016/j.eswa.2023.120462.

- Henry W. J. Reeve and Gavin Brown. Diversity and degrees of freedom in regression ensembles. *Neurocomputing*, 2018. URL <https://arxiv.org/abs/1803.00314>.
- Byron Roe. MiniBooNE particle identification. UCI Machine Learning Repository, 2005.
- Ye Tian and Yang Feng. Rase: Random subspace ensemble classification. *Journal of Machine Learning Research*, 22(45):1–93, 2021. URL <https://arxiv.org/abs/2006.08855>.
- Daniel Whiteson. HIGGS. UCI Machine Learning Repository, 2014.
- William Wolberg, Olvi Mangasarian, Nick Street, and W. Street. Breast cancer wisconsin (diagnostic), 1993. URL <https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>.
- I-Cheng Yeh. Default of credit card clients. UCI Machine Learning Repository, 2009.
- Yi Zhang, Samuel Burer, and W. Nick Street. Ensemble pruning via semi-definite programming. *Journal of Machine Learning Research*, 7:1315–1338, 2006. URL <https://jmlr.org/papers/volume7/zhang06a/zhang06a.pdf>.